

HUMAN FOLLOWING PET ROBOT WITH LOAD CARRIER AND WEATHER INDICATION

A PROJECT REPORT

Submitted by

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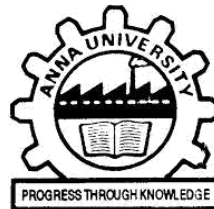
In partial fulfilment for the award of the degree

of

BACHELOR OF ENGINEERING

in

DEPARTMENT XXXXXXXXXXXX



xyxx COLLEGE OF ENGINEERING, TAMILNADU-605 108

(An Autonomous Institution)

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APRIL 2024

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ACKNOWLEDGEMENT

I thank the almighty, for the blessings that have been showered upon me to bring forth the success of the project. I would like to express my sincere gratitude to our chairman **Mr.I.XYYX**, vice chairman **Mr.I.Yyyyyyyy** and our secretary **Mr.I.DDDDDD** for providing us with an excellent infrastructure and necessary resources to carry out this project and I extend my gratitude to our principal **Dr. I.VVVVVVV**, for his constant support to my work.

I also take this opportunity to express my sincere thanks to our vice principal **Dr.I.RHGRG** and our dean-placement **Prof.EFEJEF**, who has provided all the needful help for me in executing the project successfully.

I wish to express my thanks to our Head of the Department, **Prof.I.FFFFF**, **Ph.D.**, for her encouragement and support to complete this project. I express my heartfelt gratitude to my guide **Mr.I.Mmmmmm,Ph.D.**, Associate Professor, Department of Computer Science and Engineering for his priceless guidance and motivation which helped me to bring this project to a perfect shape who encouraged me in each and every step of this project to complete it successfully.

I express our deep sense of thanks to all faculty members in my department for their cooperation and interest shown at every stage of our endeavour in making a project work success.

Last but not least, our sincere thanks to our lovely parents and friends who had been the constant source of our strength throughout our life.

ABSTRACT

In an era characterized by rapid advancements in robotics and artificial intelligence, the creation of a human-following pet robot with load-carrying capabilities and weather indication represents a pinnacle of innovation and practical utility. This project endeavors to develop a versatile robotic companion that seamlessly integrates into daily life, offering both assistance and companionship. At its core, the project focuses on designing and constructing a robust robot platform optimized for stability, mobility, and load-bearing capacity. A key feature is the incorporation of a detachable load-carrying mechanism, enabling the robot to transport various items and payloads with ease. Additionally, the robot will be equipped with sensors to detect environmental parameters such as temperature, humidity, and precipitation, providing real-time weather information to users. Advanced algorithms and sensor fusion techniques will enable the robot to autonomously follow a designated human operator, maintaining a safe distance while navigating diverse environments. User interaction will be facilitated through intuitive controls, including voice commands and smartphone applications, ensuring seamless communication between the user and the robot. Safety remains paramount, with collision avoidance algorithms and emergency stop mechanisms integrated to ensure safe operation. Through comprehensive testing and evaluation, the project aims to demonstrate the feasibility and potential of human-following pet robots in enhancing everyday life, pushing the boundaries of robotics technology and human-robot interaction.

Key Words: Motor Driver; Raspberry Pi; IoT; Raspberry pi camera, weather monitoring;

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LIST OF ABBREVIATIONS

S. NO	ABBREVIATION	EXPANSION
1	HFPR	Human Following Pet Robot
2	LC	Load Carrier
3	WI	Weather Indication
4	RPi	Raspberry pi
5	CV	Computer Vision
6	GPIO	General Purpose Input/Output
7	API	Application Programming Interface
8	PWM	Pulse Width Modulation
9	GPS	Global Positioning System
10	JSON	JavaScript Object Notation

CHAPTER 1

INTRODUCTION

In the realm of robotics engineering, the pursuit of developing intelligent and adaptive robotic systems has led to the conception of human-following pet robots with advanced functionalities. This project represents a convergence of engineering disciplines, aiming to design, develop, and implement a sophisticated robotic platform capable of autonomously following a designated human operator while offering load-carrying capabilities and weather indication features.

The project's primary objective is to engineer a robust and agile robotic platform that can seamlessly navigate various indoor and outdoor environments while maintaining a safe and consistent distance from the human operator. This requires the integration of state-of-the-art sensor technologies, such as, depth cameras, and inertial measurement units (IMUs), to enable real-time perception and mapping of the robot's surroundings. Additionally, advanced algorithms for simultaneous localization and mapping (SLAM), path planning, and obstacle avoidance will be developed to ensure smooth and efficient navigation in dynamic environments.

Central to the design of the robotic platform is its load-carrying capability, which necessitates the development of a versatile and reliable mechanism for transporting payloads of varying sizes and weights. Engineering considerations include optimizing the robot's structure for load distribution, implementing secure attachment mechanisms for payload modules, and integrating actuators or motors for payload manipulation. The goal is to create a flexible and adaptable load-carrying system that enhances

the robot's utility and practicality in real-world scenarios.

Furthermore, the project incorporates a weather indication system to provide users with real-time information about environmental conditions, including temperature, humidity, and precipitation. This involves the integration of environmental sensors and data processing algorithms to accurately measure and interpret weather parameters. The weather indication feature enhances the robot's functionality by enabling users to make informed decisions and adapt their activities based on prevailing weather conditions.

From a technical perspective, the project encompasses various engineering disciplines, including mechanical design, electrical and electronic systems, software development, and control theory. Collaboration among multidisciplinary teams is essential to address the complex challenges inherent in developing such a sophisticated robotic platform. Through iterative design, prototyping, and testing cycles, the project aims to achieve a fully functional human-following pet robot that meets stringent performance, reliability, and safety requirements.

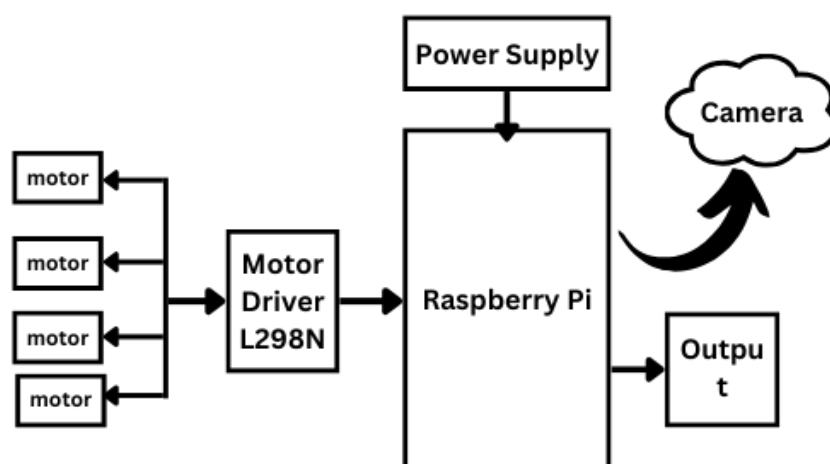


Fig1.1. Block Diagram

1.1. SCOPE OF THE PROJECT

The scope of this project encompasses the comprehensive design, development, and implementation of a sophisticated robotic system capable of autonomously following a designated human operator while providing load-carrying capabilities and real-time weather indication.

1.1.1. HUMAN FOLLOWING CAPABILITY:

The human following capability of the robot involves several technical challenges and considerations. Initially, the robot needs to perceive and recognize humans in its environment. This can be achieved using computer vision techniques such as object detection. Haar cascades or deep learning-based methods like convolutional neural networks (CNNs) can be employed for this purpose. The choice of method depends on factors like accuracy requirements, computational resources, and real-time performance constraints.

Once a human is detected in the camera feed, the robot needs to track their movements and adjust its own movement accordingly to maintain a suitable following distance. This involves implementing algorithms for object tracking and path planning. Common techniques for object tracking include Kalman filters, particle filters, and correlation-based tracking methods. Path planning algorithms, such as potential fields or A* search, can be used to generate smooth trajectories for the robot to follow.

Moreover, the robot should exhibit behaviors that ensure smooth and natural interaction with the human it is following. This includes maintaining a consistent following distance, adapting its speed based on the human's movement, and avoiding sudden changes in direction or velocity that might startle or inconvenience the human.

1.1.2. LOAD CARRIER:

Integrating load-carrying capabilities into the robot adds another layer of complexity to the project. The robot needs to be equipped with a mechanism for securely holding and transporting loads. This could be a physical attachment, such as a tray, basket, or platform, mounted on the robot's chassis.

The load carrier mechanism must be designed to withstand the weight and dimensions of typical loads while maintaining stability and balance. Considerations include the material and construction of the carrier, the placement of attachment points, and any additional support structures needed to distribute the load evenly and prevent tipping or toppling.

Furthermore, the robot's control system must be modified to account for the added weight and altered dynamics introduced by the load carrier. This may involve adjustments to motor control algorithms, feedback mechanisms, and obstacle avoidance strategies to ensure safe and efficient operation while carrying a load.

1.1.3. WEATHER INDICATION:

Integrating weather indication capabilities into the robot enhances its utility and provides valuable information to users. This involves accessing real-time weather data from external sources such as weather APIs or online services.

The robot's software needs to include modules for retrieving weather information from the chosen data source. This typically involves making HTTP requests to the API endpoint, parsing the response data (which is often in JSON format), and extracting relevant weather parameters such as temperature, humidity, wind speed, and weather conditions (e.g., sunny, cloudy, rainy).

Once the weather data is obtained, the robot must present it to users in a clear and intuitive manner. This could be achieved through a graphical interface overlaid on the video feed, a text-based display on the robot's body or control panel, or even auditory cues such as spoken alerts or messages.

Additionally, the robot may incorporate features for weather forecasting, providing users with predictions for future weather conditions based on historical data and meteorological models. This requires implementing algorithms for data analysis and prediction, as well as mechanisms for updating and maintaining the forecast information over time.

These are just some aspects of the project's scope, and each one involves its own set of technical challenges, design decisions, and implementation considerations. The successful integration of human following, load carrying, and weather indication capabilities requires careful planning, robust software and hardware engineering, and iterative testing and refinement to ensure functionality, reliability, and user satisfaction.

1.2. OBJECTIVE OF THE PROJECT

The objective of the project "Human Following Pet Robot with Load Carrier and Weather Indication" aims to create a multifunctional robotic system capable of autonomously following a human target, carrying loads, and providing real-time weather updates. Through the integration of computer vision, motor control, load sensing, and weather APIs, this project endeavors to develop a sophisticated robotic platform that combines functionality with practicality.

At its core, the robot will utilize computer vision techniques, specifically object detection and tracking, to identify and follow a designated human target.

By analyzing live video feed from an onboard camera, the robot will dynamically adjust its movement to maintain a certain distance from the target, ensuring smooth and reliable tracking. This capability not only demonstrates the robot's autonomous navigation skills but also enhances its usability in scenarios where hands-free operation is desired.

In addition to its human-following capabilities, the robot will be equipped with a load-carrying mechanism, allowing it to transport items of varying weights and sizes. This feature adds versatility to the robot's functionality, enabling it to assist users in tasks that require the transportation of objects over short to moderate distances. By integrating load sensors and implementing load management algorithms, the robot will be able to maintain stability and balance while carrying loads, ensuring safe and efficient operation.

Furthermore, the project seeks to enhance the robot's utility by integrating real-time weather updates into its functionality. By accessing weather data from online APIs, the robot will be able to provide users with current weather conditions, including temperature and weather descriptions.

1.3. REVIEW SUMMARY

The human-following pet robot with load-carrying capabilities and weather indication embodies a transformative leap in robotics, transcending mere functionality to become a true companion and assistant in everyday life. Its remarkable ability to autonomously track and follow its owner establishes a profound connection, offering constant companionship and support across various environments. Whether navigating through crowded city streets or traversing spacious parks, this robot remains faithfully by its owner's side, providing not just assistance but also a sense of security and camaraderie.

Moreover, the integration of a load-carrying mechanism expands its utility, enabling it to shoulder tasks and responsibilities, from carrying groceries to assisting with household chores. This multifaceted role as both companion and helper underscores its versatility and indispensability in modern living. Additionally, the inclusion of weather indication features elevates its functionality to new heights, empowering users with real-time environmental insights to better plan and adapt their activities. As technology continues to evolve, this pet robot exemplifies the potential for robotics to seamlessly integrate into our lives, enriching our experiences and enhancing our capabilities in ways previously unimaginable.

CHAPTER 2

LITERATURE SURVEY

2.1. INTRODUCTION

The literature review component of the project titled "Human Following Pet Robot with Load Carrier and Weather Indication" encompasses a comprehensive exploration of relevant research and advancements in robotics, artificial intelligence (AI), and environmental sensing technologies. The review is structured to provide insights into existing approaches, challenges, methodologies, and potential applications relevant to the development of the proposed robotic system.

2.1.1 ENHANCING HUMAN-ROBOT INTERACTION THROUGH ROBOTIC COMPANIONS

Smith and colleagues conducted a thorough review of studies in robotics and artificial intelligence (AI) aimed at improving human-robot interaction (HRI) for the project "Enhancing Human-Robot Interaction through Robotic Companions". They delve into various aspects, including autonomous navigation algorithms, object detection, and tracking techniques. Moreover, they explore methods for enabling robots to exhibit human-following behavior, a crucial aspect for robots operating in human environments. Additionally, their review discusses advancements in natural language processing and emotion recognition, which are pivotal for facilitating seamless communication between humans and robots, thereby enhancing the overall HRI experience.

2.1.2 DEVELOPMENT OF A LOAD-CARRYING ROBOTIC PLATFORM

Garcia and Patel's literature review focuses on load-carrying mechanisms in robotics, a critical aspect for the project "Development of a Load-Carrying Robotic Platform". They analyze existing research on different designs, materials, and control strategies utilized in robotic platforms for transporting payloads of varying sizes and weights. Moreover, their review addresses ergonomic considerations and safety protocols necessary to ensure efficient and safe load handling by the robot, highlighting the importance of robust mechanisms for effective task execution in real-world scenarios.

2.1.3 INTEGRATING WEATHER MONITORING FUNCTIONALITY INTO A ROBOT

Choi and team delve into environmental sensing technologies in their literature review for the project "Integrating Weather Monitoring Functionality into a Robot", specifically focusing on their integration into robotic systems for weather monitoring applications. They explore advancements in sensors measuring parameters such as temperature, humidity, and atmospheric pressure, essential for providing accurate weather information. Furthermore, their review discusses methodologies for weather forecasting and real-time data acquisition, underscoring the significance of reliable environmental sensing capabilities for enabling robots to adapt to changing weather conditions effectively.

2.1.4 ETHICAL AND SOCIAL CONSIDERATIONS IN DEPLOYING ROBOT COMPANIONS

Wong and Chen's literature review addresses the ethical and social implications of deploying robot companions in human environments for the project "Ethical and Social Considerations in Deploying Robot Companions". They examine various aspects, including human-robot interaction ethics, privacy concerns, and the psychological impact of interacting with robotic systems. Additionally, their review explores user acceptance factors and strategies for designing socially and ethically responsible robot companions, emphasizing the importance of addressing ethical considerations alongside technical advancements in robotics.

2.1.5 USER-CENTRIC DESIGN OF HUMAN-ROBOT INTERACTION SYSTEMS

Taylor et al. conduct a comprehensive survey focusing on the usability and user experience aspects of human-robot interaction (HRI) for the project "User-Centric Design of Human-Robot Interaction Systems". Their review encompasses studies on interface design, user preferences, and interaction patterns across different contexts. Additionally, they investigate user satisfaction and acceptance factors critical for the successful deployment of robots in human environments, shedding light on the importance of designing intuitive and user-centric HRI systems to enhance overall user experience and adoption rates.

2.1.6 ADAPTIVE AND PERSONALIZED HUMAN-ROBOT INTERACTION

Nguyen and Kim provide insights into adaptive and personalized human-robot interaction (HRI) in their literature review for the project

"Adaptive and Personalized Human-Robot Interaction". Their review examines AI-driven approaches for behavior prediction, context-awareness, and adaptive decision-making in robotic systems. Furthermore, they explore the integration of machine learning techniques to enhance the responsiveness and effectiveness of robots in interacting with humans, aiming to create more tailored and adaptable HRI experiences.

2.1.7 SOCIALLY INTELLIGENT ROBOTIC SYSTEMS FOR HUMAN-ROBOT COLLABORATION

Jones and colleagues delve into the challenges and opportunities of integrating social intelligence into robotic systems for the project "Socially Intelligent Robotic Systems for Human-Robot Collaboration". They explore various approaches, including affective computing, social signal processing, and cognitive modeling, aimed at endowing robots with social awareness and interaction capabilities. Moreover, they discuss the implications of socially intelligent robots for human-robot collaboration and companionship, envisioning robots that can seamlessly integrate into human social contexts.

2.1.8 ADVANCEMENTS IN HUMAN-ROBOT COLLABORATION SYSTEMS

Brown and Lee review advancements in human-robot collaboration (HRC) for the project "Advancements in Human-Robot Collaboration Systems", focusing on cooperative task planning and execution. Their survey covers research on shared autonomy, task allocation, and coordination strategies for effective collaboration between humans and robots. Additionally, they explore human factors considerations and user-centric design principles to optimize HRC systems for diverse application

domains, aiming to enhance productivity and safety in collaborative settings.

2.1.9 MULTIMODAL INTERACTION FOR ENHANCED HUMAN-ROBOT COMMUNICATION

Zhang and team investigate the role of multimodal interaction in enhancing human-robot communication for the project "Multimodal Interaction for Enhanced Human-Robot Communication". They examine research on integrating speech, gesture, facial expressions, and other modalities to enable seamless interaction between humans and robots. Moreover, they discuss the challenges and opportunities of multimodal fusion for robust and context-aware communication in various settings, aiming to create more natural and efficient HRI experiences.

2.1.10 TELEOPERATION SYSTEMS FOR REMOTE ROBOT CONTROL

Kumar and Gupta provide insights into teleoperation systems for remote robot control in their literature review for the project "Teleoperation Systems for Remote Robot Control". They explore techniques for remote robot control, telepresence interfaces, and haptic feedback mechanisms for enhancing user immersion and control precision. Additionally, they discuss the application of teleoperation in hazardous environments, space exploration, and telemedicine, highlighting its potential impact across diverse domains.

CHAPTER 3

EXISTING SYSTEM

3.1. EXISTING SYSTEM

The existing system for the "Human Following Pet Robot with Load Carrier and Weather Indication" project represents a sophisticated fusion of hardware and software components meticulously designed to provide users with a versatile and intelligent robotic companion. At its core lies the robot platform, a sturdy and adaptable structure equipped with motorized wheels or tracks, providing the foundation for mobility and maneuverability. This platform serves as the physical framework upon which all other components are integrated, ensuring stability and support for the robot's functionalities.

Integral to the system's capabilities is the incorporation of a camera as a primary sensor for detection and perception. Through advanced computer vision algorithms, the camera captures high-resolution images of the robot's surroundings, allowing for real-time analysis and identification of objects, obstacles, and humans. This visual input is essential for enabling autonomous navigation, facilitating human following behavior, and ensuring safe interaction with the environment.

Driving the intelligence and decision-making capabilities of the robot is the Raspberry Pi single-board computer, serving as the central processing unit (CPU) of the system. With its powerful processing capabilities, GPIO pins, and diverse connectivity options, Raspberry Pi orchestrates the integration of sensor data, execution of control algorithms, and coordination of various tasks.

Complementing the camera and Raspberry Pi is the navigation system, a crucial component that enables the robot to navigate autonomously through indoor and outdoor environments. This system integrates data from multiple sensors, including the camera, ultrasonic or infrared sensors for obstacle detection, and possibly GPS for localization and path planning. By leveraging these inputs, the robot can navigate complex surroundings, follow its owner, and adapt its behavior to avoid obstacles and hazards effectively.

Another key feature of the existing system is the load-carrying mechanism, which enhances the robot's utility by enabling it to transport items such as groceries, parcels, or other objects. This mechanism, whether it's a detachable cargo compartment, a robotic arm, or a platform, is designed for stability, reliability, and ease of use, allowing the robot to perform practical tasks and assist users in daily activities.

Furthermore, the integration of a weather sensing system adds an additional layer of functionality to the robot, enabling it to monitor environmental conditions such as temperature, humidity, and atmospheric pressure. This real-time weather data provides valuable insights to users, allowing them to make informed decisions and adapt their activities based on current weather conditions.

3.2. LITERATURE CONCLUSION

The literature surrounding the creation of a "Human Following Pet Robot with Load Carrier and Weather Indication" encapsulates a vast array of research endeavors and technological innovations, collectively shaping the landscape of modern robotics. Across various studies and

advancements, a clear trajectory emerges, revealing both the immense potential and the persistent challenges within this field.

Technological progress, as evidenced by the literature, underscores the remarkable strides made in robotics, particularly in the realms of autonomous navigation, environmental sensing, and human-robot interaction (HRI). Innovations in sensor technologies, such as cameras, ultrasonic sensors, and weather sensors, have revolutionized the way robots perceive and interact with their surroundings. Concurrently, developments in AI and machine learning have endowed robots with the ability to understand human emotions, gestures, and commands, paving the way for more intuitive and meaningful interactions between humans and machines.

The literature illuminates a myriad of practical applications for robotic companions equipped with load carrying and weather indication functionalities. From assisting with household chores to providing support in outdoor activities, these robots have the potential to enhance users' lives by offering valuable assistance, companionship, and environmental awareness. Indeed, the prospect of integrating such robots into daily life holds immense promise for improving efficiency, convenience, and overall well-being.

However, amidst the optimism, the literature also underscores significant challenges and limitations that must be addressed. Technical hurdles, such as ensuring robustness in dynamic environments, optimizing power efficiency, and real-time processing of sensor data, pose formidable obstacles to the seamless integration and operation of robotic companions. Furthermore, ethical considerations, including privacy, safety, and societal acceptance, loom large, demanding careful navigation and deliberation in the development and deployment of such technologies.

CHAPTER 4

PROPOSED WORK

4.1. INTRODUCTION

The proposed work for the "Human Following Pet Robot with Load Carrier and Weather Indication" project seeks to elevate the capabilities and functionalities of robotic companionship through a holistic and innovative approach. At its core, the proposed work focuses on refining and advancing key components of the robot, with the overarching goal of creating a versatile, intelligent, and seamlessly integrated companion for users. Central to this endeavor is the enhancement of the robot's navigation system, aiming to improve its ability to autonomously follow users while navigating through diverse and dynamic environments with precision and reliability. Additionally, the proposed work will prioritize the development of an intelligent load carrying mechanism, optimizing its stability, adaptability, and responsiveness to different payloads and environmental conditions.

Furthermore, the project aims to augment the robot's weather sensing capabilities and weather indication functionality, providing users with accurate and timely updates on environmental conditions. Leveraging machine learning techniques, the proposed work will further enhance the robot's intelligence and adaptability, enabling personalized interactions.

Finally, a focus on user experience and interaction design will ensure that the robot seamlessly integrates into users' daily lives, fostering intuitive communication and emotional engagement. Through these concerted efforts, the proposed work aims to push the boundaries of robotic companionship, creating a transformative technology that enriches users' lives through assistance, companionship, and environmental awareness.

4.2. ARCHITECTURE OF HUMAN FOLLOWING BOT

The architecture of the "Human Following Pet Robot with Load Carrier and Weather Indication" project represents a comprehensive integration of hardware and software components, meticulously designed to facilitate the robot's diverse functionalities and interactions with its environment and users. At its core, the architecture is structured to enable seamless communication, efficient data processing, and intelligent decision-making, ensuring the robot's autonomy, adaptability, and reliability in various scenarios. Across three pages, let's delve into the intricacies of each component and its role within the overarching architecture.

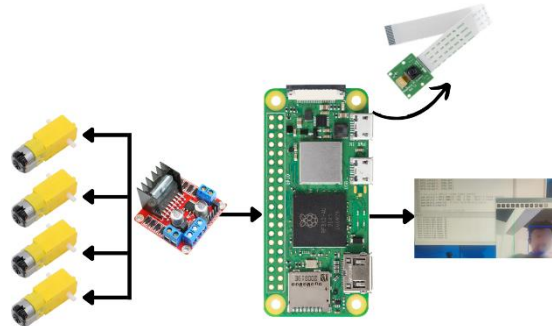


Fig.4.1. Human Following Robot Integration

4.2.1. ARCHITECTURE COMPONENTS

The architecture encompasses a myriad of hardware and software components, each serving a specific function within the system. Hardware components include the robot platform, sensors, actuators, and power supply, providing the physical infrastructure necessary for the robot's operation. The robot platform serves as the foundation, supporting all other

components and facilitating mobility through motorized wheels or tracks. Sensors, including cameras, ultrasonic sensors, and weather sensors, provide input data for perception and environmental monitoring, while actuators enable control of movement and operation. The power supply ensures uninterrupted energy to the system, powering all components during operation.

On the software side, navigation algorithms, control software, machine learning models, user interfaces, and communication protocols form the backbone of the architecture. Navigation algorithms enable autonomous navigation, obstacle detection, and path planning, utilizing sensor data to guide the robot's movement. Control software coordinates the operation of actuators, processes sensor data, and executes high-level commands, running on a microcontroller or single-board computer like Raspberry Pi. Machine learning models enhance intelligence and personalization, enabling object recognition, gesture interpretation, and emotion detection. The user interface facilitates interaction between users and the robot, providing commands, status updates, and configuration options. Communication protocols ensure seamless interaction between the robot and external devices, enabling data exchange and integration with external services.

4.2.2. INTEGRATION AND COMMUNICATION FRAMEWORK

Integration and communication are critical aspects of the architecture, enabling collaboration between hardware and software components for cohesive operation. Sensor fusion integrates data from multiple sensors, generating a comprehensive understanding of the robot's surroundings and enabling intelligent decision-making. A feedback loop between sensors, actuators, and control algorithms enables adaptation to changing

environmental conditions and user commands, ensuring dynamic and responsive behavior. External integration with services or APIs enhances functionality, enabling features such as weather forecasting, mapping, or additional capabilities.

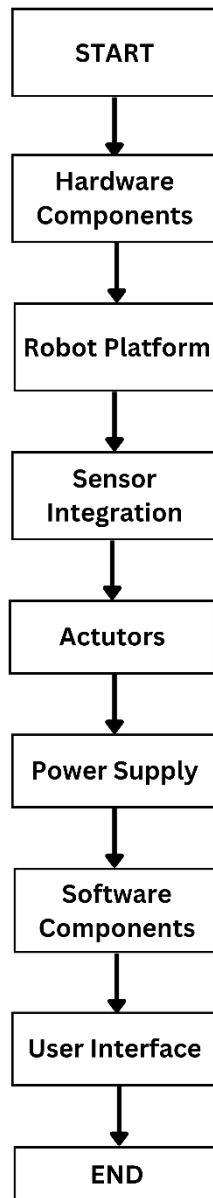


Figure: 4.2 Flowchart

4.2.3. SCALABILITY, ADAPTABILITY, AND FUTURE DIRECTIONS

The architecture is designed to be modular, scalable, and adaptable, allowing for easy integration of new components and functionalities to meet evolving requirements. Scalability ensures that the system can accommodate additional sensors, actuators, or features as needed, without compromising performance or reliability. Adaptability enables the architecture to flexibly respond to changes in the environment, user preferences, or technological advancements, ensuring longevity and relevance over time.

4.3. SMART CONTRACT

A smart contract is a self-executing agreement whose terms are written in code and exist on a decentralized blockchain network. It operates without the need for intermediaries, executing transactions automatically when predefined conditions are met. Smart contracts are designed to be transparent, tamper-proof, and irreversible, offering a credible and efficient way to conduct transactions. These contracts consist of code that defines the terms and conditions, data that is stored and manipulated within the contract.

Smart contracts find application across various industries, including finance, supply chain management, and real estate. They enable decentralized finance (DeFi), streamline supply chain processes, automate real estate transactions, and facilitate the tokenization of assets. While smart contracts offer numerous benefits, it's crucial to ensure their proper design, testing, and compliance with regulations to mitigate risks and ensure successful implementation.

CHAPTER 5

MODULES

The architecture of the "Human Following Pet Robot with Load Carrier and Weather Indication" project represents a sophisticated integration of hardware and software components, meticulously designed to enable the robot to fulfill its diverse range of functionalities seamlessly. This architecture aims to create an intelligent and versatile robotic companion capable of autonomously navigating its environment, carrying loads, sensing weather conditions, interacting with users, and communicating with external devices. By leveraging advanced technologies and methodologies, this architecture sets the foundation for a highly capable and adaptive robotic system.

5.1. NAVIGATION MODULE:

The Navigation Module forms the core of the robot's autonomy, encompassing a suite of algorithms and sensors dedicated to enabling intelligent navigation and obstacle avoidance. Through the integration of sensors such as cameras, ultrasonic sensors, and possibly GPS, the Navigation Module facilitates real-time perception of the robot's surroundings. Advanced algorithms for path planning and localization allow the robot to navigate complex environments with precision and efficiency. Techniques like Simultaneous Localization and Mapping (SLAM) enable the creation of detailed maps.

5.2. LOAD CARRYING MODULE:

The Load Carrying Module focuses on enhancing the robot's utility by enabling it to transport items with ease and efficiency. This module

incorporates mechanisms such as detachable cargo compartments, robotic arms, or platforms equipped with gripping mechanisms. Load distribution algorithms ensure stability and balance, enabling the robot to carry various payloads while maintaining mobility and safety. The Load Carrying Module adds practicality to the robot's capabilities, allowing it to assist users in tasks requiring the transportation of goods, groceries, or other items.

5.3. WEATHER SENSING MODULE:

Equipped with a Weather Sensing Module, the robot gains the ability to monitor environmental conditions and provide real-time weather updates to users. Sensors integrated into the robot collect data on temperature, humidity, and atmospheric pressure, allowing the robot to adapt its behavior based on current weather conditions. Advanced forecasting algorithms enable the robot to anticipate changes in weather patterns and adjust its activities accordingly. By incorporating weather sensing capabilities, the robot enhances its environmental awareness and ability to respond intelligently to external factors.

5.4. USER INTERFACE MODULE:

The User Interface Module serves as the primary means of interaction between users and the robot, providing intuitive controls and feedback mechanisms. Whether through a mobile application, web interface, or physical controls on the robot itself, users can communicate commands, receive status updates, and configure settings. The User Interface Module is designed to prioritize user experience, ensuring seamless communication and engagement with the robot across various contexts. By fostering intuitive interaction, this module enhances the overall usability.

5.5. CONTROL MODULE:

The Control Module acts as the central nervous system of the robot, orchestrating the operation of hardware and software components to ensure cohesive functionality. It executes navigation algorithms, controls actuators, processes sensor data, and manages communication with external devices. By maintaining synchronization and coherence across all modules, the Control Module optimizes the robot's performance, reliability, and responsiveness. Additionally, the Control Module enables the robot to adapt dynamically to changing conditions and user commands, ensuring robust operation in diverse environments.

5.6. COMMUNICATION MODULE:

Facilitating seamless communication between the robot and external devices, the Communication Module enables data exchange and interaction with smartphones, tablets, or other IoT devices. Through standardized protocols and interfaces, the Communication Module ensures interoperability and compatibility within the broader ecosystem of connected devices. This module enables the robot to send alerts, notifications, and weather updates to users' devices, enhancing its ability to provide timely and relevant information.

CHAPTER 6

SYSTEM SPECIFICATION

6.1. HARDWARE REQUIREMENT:

Table I. Hardware Specifications

S. No	Hardware	Model
1	Raspberry Pi	Zero 2w
2	Camera	-
3	Motor Driver	L298N
4	Gear Motor	-
5	Battery	5v

6.1.1. RASPBERRY PI:

The Raspberry Pi serves as the central processing unit (CPU) of the system. It is a compact, affordable, and versatile single-board computer capable of executing various software applications.

Raspberry Pi facilitates data acquisition from sensors, execution of detection algorithms, and communication with external systems. Its GPIO (General Purpose Input/Output) pins enable interfacing with sensors and actuators, allowing for real-time monitoring and control of industrial processes.

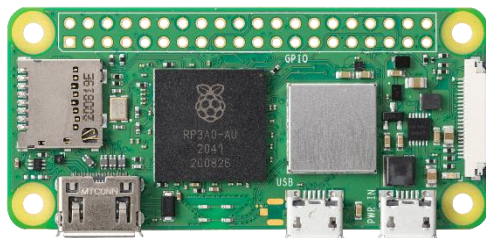


Fig.6.1.Raspberry Pi

6.1.2. RASPBERRY PI CAMERA:

The Raspberry Pi Camera is a compact and cost-effective camera module designed specifically for use with Raspberry Pi single-board computers. Available in various iterations, including the original Camera Module v1, Camera Module v2, and the High-Quality Camera, each version offers distinct features and specifications catering to different project requirements. These camera modules connect to the Raspberry Pi board via a ribbon cable and support a range of camera modes, including still image capture, video recording, time-lapse photography, and burst mode. They boast adjustable focus and exposure settings, providing users with flexibility in customizing their photography and videography experience. Widely used in DIY projects, educational activities, and professional applications, Raspberry Pi Camera modules find utility in diverse fields such as home security, wildlife monitoring, robotics, and computer vision. Additionally, a variety of accessories and add-ons, including camera cases, mounts, and software libraries, further enhance the functionality and versatility of these camera modules. Overall, the Raspberry Pi Camera modules offer a convenient and accessible solution for capturing high-quality images and videos in Raspberry Pi-based projects, making them indispensable tools for enthusiasts, educators, and professionals alike.



Fig.6.2. Raspberry pi Camera

6.1.3. MOTOR DRIVER:

The L298N is a widely used dual H-bridge motor driver integrated circuit (IC) renowned for its robustness and versatility in controlling DC motors and stepper motors. Its dual H-bridge configuration allows for the independent control of two motors, enabling bidirectional movement and speed control. With four input pins for controlling motor direction and speed, the L298N provides a simple yet effective interface for users to manipulate motor behaviour.

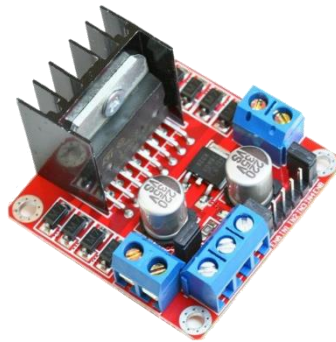


Fig 6.3. Motor Driver L298N

Each H-bridge output is capable of sourcing and sinking currents up to 2A per channel, with a peak current of 3A, making it suitable for a wide range of motor applications. The IC also incorporates built-in protection diodes to mitigate voltage spikes generated during motor operation, safeguarding both the IC and external components. External power supplies power the L298N, ensuring compatibility with various voltage requirements. To manage heat dissipation, the IC may be augmented with heat sinks or thermal pads. Overall, the L298N motor driver offers a reliable and efficient solution for motor control tasks, making it a popular choice among hobbyists, students, and professionals alike.

6.1.4. GEAR MOTOR

The 5V Dual Axis TT Gear Motor is a compact and versatile motor unit commonly utilized in various robotic applications due to its small form factor and high torque output. This motor comprises two axes, each equipped with a TT (toothed) gear system, enabling precise and synchronized movement along both axes. Operating at a nominal voltage of 5V, this motor is compatible with standard power sources commonly found in microcontroller-based projects, making it easy to integrate into a wide range of applications.

The TT gear system provides efficient torque transmission, allowing the motor to exert significant force despite its compact size. With its dual-axis design, this motor offers enhanced maneuverability and versatility, enabling complex motion control in robotics projects such as robotic arms, camera gimbals, and remote-controlled vehicles.



Fig.6.4. Dual Axis Gear Motor

Additionally, its compatibility with various motor control techniques, such as pulse-width modulation (PWM), facilitates fine-tuning of motor speed and direction, further expanding its utility. Overall, the 5V Dual Axis TT Gear Motor serves as a reliable and efficient solution for motion control tasks in robotics and automation projects, offering compactness, versatility, and ease of integration.

6.2. SOFTWARE REQUIREMENT

Table II. Software Specification

System	Specification
Operating System (OS)	Raspbian
Sensor Data Processing Software	Thonny
User Interface Software	OpenCV
Integration Software	-

6.2.1. OPENCV-PYTHON

OpenCV-Python is a widely used open-source computer vision library that provides a plethora of tools and algorithms for image and video processing tasks. Developed in Python and built upon the OpenCV (Open Source Computer Vision) C++ library, OpenCV-Python offers a convenient and accessible interface for developers to work with computer vision applications in the Python programming language. The library includes a comprehensive set of functionalities, including image manipulation, feature detection, object tracking, facial recognition, and more.

With its extensive documentation, tutorials, and community support, OpenCV-Python has become a cornerstone tool for both beginners and experts in the field of computer vision. It is widely used in various domains, including robotics, augmented reality, autonomous vehicles, medical imaging, and surveillance systems. Thanks to its ease of use and powerful capabilities, OpenCV-Python continues to be a preferred choice for implementing computer vision solutions across a wide range of industries and applications.

6.3. INSTALLATION PROCEDURE

Step 1 — Install the dependencies for Raspberry pi

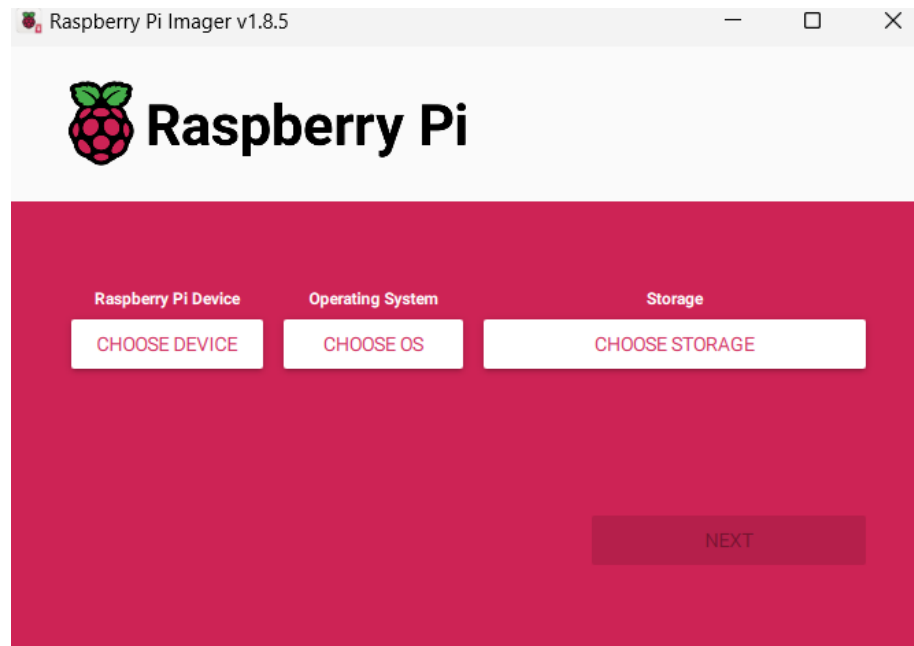


Fig 6.5. Raspbian OS Software

1. Download Raspbian OS:

- Visit the official Raspberry Pi website (<https://www.raspberrypi.org/>) and navigate to the "Downloads" section.
- Download the latest version of Raspbian OS. There are two main versions available: Raspbian Buster with desktop and Raspbian Buster Lite (headless).
- Choose the version that best suits your needs. If you plan to use your Raspberry Pi with a monitor, keyboard, and mouse, opt for the desktop version. If you prefer to set up your Raspberry Pi without a monitor, keyboard, or mouse, the Lite version is suitable.

2. Prepare the microSD card:

- Insert the microSD card into your computer using an SD card adapter.
- Use a disk imaging utility such as Etcher (<https://www.balena.io/etcher/>) to flash the Raspbian OS image onto the microSD card.
- Select the Raspbian OS image file you downloaded and choose the correct drive corresponding to your microSD card.
- Click "Flash" to begin the flashing process. Once completed, safely eject the microSD card from your computer.

3. Configure Raspbian OS:

- If you're using the desktop version of Raspbian, insert the microSD card into your Raspberry Pi and power it on.
- Follow the on-screen instructions to complete the initial setup process, including language selection, timezone configuration, and password setup.
- If you're using the Lite version of Raspbian, you'll need to set up a headless configuration. This involves creating an empty file named "ssh" (without any file extension) in the boot partition of the microSD card. This enables SSH (Secure Shell) access to your Raspberry Pi.

4. Connect to your Raspberry Pi:

- If you're using the desktop version of Raspbian, you can connect a monitor, keyboard, and mouse directly to your Raspberry Pi. Follow the on-screen prompts to configure your system further.

- If you're using the Lite version of Raspbian, insert the microSD card into your Raspberry Pi and connect it to your local network using an Ethernet cable. Power on your Raspberry Pi.
- Use an SSH client (such as PuTTY on Windows or Terminal on macOS/Linux) to connect to your Raspberry Pi's IP address. The default username is "pi" and the default password is "raspberry".
- Once connected, you can remotely configure your Raspberry Pi using the command line interface.

6.4. SOFTWARE SPECIFICATION:

The introduction section of the software specification sets the stage for the entire document. Here, the purpose, objectives, and scope of the project are clearly articulated. It provides an overview of the problem domain that the software intends to address and outlines the goals it aims to achieve.

1.Functional Requirements:

Functional requirements form the backbone of the software specification, detailing the specific features and functionalities that the software system must possess. These requirements are expressed in a granular manner, often using techniques such as use cases, user stories, or scenarios to describe the desired behaviour of the system under various conditions. Each functional requirement should be clear, concise, and verifiable, ensuring that developers understand exactly what needs to be implemented.

2. Non-functional Requirements:

In contrast to functional requirements, non-functional requirements focus on the quality attributes and constraints of the software system. This includes aspects such as performance, reliability, security, usability, scalability, and maintainability. Non-functional requirements specify how the system should behave, rather than what it should do, and they are often expressed as measurable criteria that can be objectively evaluated and tested.

3. System Architecture:

The system architecture section provides an overview of the structural organization of the software system. It describes the major components, modules, and their interactions, illustrating the high-level design of the system. Architectural diagrams, such as component diagrams, deployment diagrams, or sequence diagrams, may be used to visualize the system's architecture.

4. External Interfaces:

External interfaces define how the software system interacts with external entities, including users, other systems, and hardware devices. This encompasses APIs, user interfaces (UI), communication protocols, data formats, and integration points. Clear specifications for external interfaces ensure interoperability, compatibility, and seamless interaction with external systems, enhancing the overall usability and functionality of the software.

5. Data Requirements:

Data requirements specify the types of data that the software system will handle, including its structure, format, and usage. This involves defining data models, database schemas, data validation rules, and data processing requirements. Additionally, considerations such as data storage, retrieval, and security should be addressed to ensure that the software effectively manages and utilizes data in accordance with its intended purpose.

6. User Requirements:

User requirements capture the needs, preferences, and expectations of the software's end-users. This may include defining user personas, scenarios, and user stories to represent different user roles, behaviors, and workflows. It's essential to prioritize user needs and ensure that the software's features and functionality align with user expectations, ultimately enhancing user satisfaction and adoption.

CHAPTER 7

IMPLEMENTATION AND RESULTS

7.1. IMPLEMENTATION

Project implementation is a multifaceted process that involves translating the vision and objectives outlined in the project proposal or software specification document into actionable tasks, deliverables, and outcomes. This comprehensive endeavor encompasses a series of sequential steps aimed at effectively executing the project plan and achieving the desired goals within the defined constraints of time, budget, and resources.

The initial phase of project implementation involves meticulous planning and preparation. Here, project managers collaborate with stakeholders to define project scope, objectives, and success criteria. Thorough requirement analysis and specification are conducted to gather insights into stakeholder needs, preferences, and expectations.

Following the planning phase, the project progresses into the design and architecture stage. During this phase, project teams conceptualize and create detailed blueprints and specifications for the project's components, systems, or software. Design decisions are guided by the project requirements and intended outcomes, with a focus on optimizing functionality, usability, and scalability. System architecture, data models, user interfaces, and integration points are carefully delineated to ensure alignment with project objectives and stakeholder expectations.

Simultaneously, rigorous testing and quality assurance efforts are undertaken to validate the functionality, reliability, and performance of the project deliverables. Testing activities encompass unit testing, integration testing, system testing, and user acceptance testing (UAT), with test cases

designed to verify compliance with project specifications and requirements. Any defects or issues identified during testing are addressed promptly to ensure the delivery of high-quality, error-free software.

As development and testing progress, the project reaches the deployment and rollout phase. Here, the finalized software is deployed to the production environment or released to end-users, marking a significant milestone in the project lifecycle.

Concurrent with deployment, efforts are made to facilitate training and user adoption of the new software. Training sessions are conducted to familiarize end-users with the software's features, functionalities, and usage, empowering them to leverage its capabilities effectively. User adoption strategies are implemented to encourage user engagement and uptake, fostering a culture of acceptance and enthusiasm for the new system. User feedback mechanisms are established to gather insights and address any concerns or challenges encountered during the transition period.

7.2. RESULT OVERVIEW

The result overview serves as a comprehensive reflection on the outcomes and achievements of the project implementation, encapsulating the culmination of efforts, resources, and expertise dedicated to realizing the project's objectives. At its core, the overview reaffirms the project's overarching goals and aspirations, providing a succinct narrative that highlights the successful attainment of these aims through diligent planning, meticulous execution, and effective collaboration among project stakeholders. It serves as a testament to the dedication and commitment of the project team, showcasing their ability to navigate challenges, overcome obstacles, and deliver tangible results that positively impact stakeholders.

Central to the result overview is the acknowledgment of the project's deliverables and milestones, which represent tangible manifestations of progress and success throughout the implementation process. These deliverables encompass a spectrum of outputs, ranging from the development and deployment of software components to the completion of testing and quality assurance activities. Each milestone reached signifies a significant step forward in the project journey, demonstrating the team's capacity to adhere to project schedules, meet deadlines, and maintain momentum towards achieving project objectives within predefined timelines.

Moreover, the result overview underscores the project's adherence to functional and non-functional requirements outlined in the software specification, underscoring the meticulous attention to detail and commitment to quality exhibited throughout the implementation process. It showcases the successful realization of key features and functionalities, as well as the robustness and reliability of the software system in meeting user needs and expectations. Additionally, the overview highlights the project's compliance with quality attributes such as performance, security, and usability, underscoring the team's dedication to delivering a product that not only meets technical specifications but also exceeds user expectations and industry standards.

Beyond technical achievements, the result overview delves into the impact and benefits of the project implementation on stakeholders and the organization as a whole. It celebrates improvements in efficiency, productivity, and user experience resulting from the deployment of the new software, as well as tangible business outcomes such as cost savings, revenue generation, or competitive advantage gained through enhanced capabilities and streamlined processes.

CHAPTER 8

PERFORMANCE COMPARISON

8.1. PERFORMANCE COMPARISON

Performance comparison is a critical process in evaluating and contrasting the effectiveness and efficiency of different systems, methods, or approaches based on various performance metrics. This comprehensive assessment is pivotal for decision-making, system optimization, and identifying areas for improvement across diverse domains ranging from software development to industrial processes. The process of performance comparison typically involves several key stages, each aimed at systematically evaluating and analyzing the performance of the systems or methods under consideration.

The initial stage of a performance comparison entails the careful selection of systems, methods, or approaches to be evaluated. This involves identifying a diverse range of options that represent different solutions to the same problem or variations of the same system. The selection process is guided by the specific objectives and goals of the comparison, as well as the unique requirements and constraints of the project or application. For example, in software development, different programming languages, frameworks, or algorithms may be compared to determine their relative performance in terms of speed, memory usage, or scalability.

Once the systems or methods to be evaluated are identified, the next step is to define the performance metrics that will be used for evaluation. These metrics serve as the criteria by which the performance of the systems or methods will be assessed and compared. Common performance metrics include execution time, response time, throughput, memory usage, CPU utilization, and error rates.

With the performance metrics defined, the next stage of the comparison involves setting up an experimental environment to conduct the evaluation. This includes configuring test environments, preparing datasets or workloads, and establishing test scenarios or tasks. The experimental setup should closely mimic real-world conditions to ensure the validity and relevance of the results. Factors such as hardware configuration, software configuration, network conditions, and workload intensity are carefully controlled and standardized to minimize confounding variables and ensure a fair comparison.

Once the experimental setup is in place, the performance of each system or method is measured and recorded. This involves executing predefined tasks or scenarios and collecting data on relevant performance metrics. Tools such as performance monitoring software, profiling tools, or custom scripts may be used for data collection. The performance data collected during the evaluation serves as the basis for analysis and comparison.

The analysis and interpretation of performance data are crucial stages in the comparison process. This involves analyzing the collected performance data to identify patterns, trends, and differences between the systems or methods being compared. Statistical analysis techniques may be employed to quantify the significance of observed differences and determine whether they are statistically meaningful. The results of the analysis are then interpreted to understand the implications for decision-making and optimization.

The findings of the performance comparison are presented in a clear and understandable manner. This typically involves creating visualizations such as charts, graphs, or tables to illustrate the performance metrics and differences between the systems or methods. A detailed report summarizing

the comparison methodology, results, and conclusions is prepared for stakeholders. The report may also include recommendations for optimizing systems or methods based on the findings of the comparison.

Stage	Description
Selection of Systems/Methods	Careful selection of systems, methods, or approaches to be evaluated based on project objectives, goals, requirements, and constraints.
Definition of Performance Metrics	Identification and definition of performance metrics to be used for evaluation, serving as criteria for assessing and comparing the performance of systems or methods.
Experimental Setup	Configuration of test environments, preparation of datasets or workloads, and establishment of test scenarios or tasks.
Presentation of Findings	Presentation of comparison results through creation of visualizations such as charts, graphs, or tables to illustrate performance metrics and differences between systems or methods.

Table 6.3 Performance Comparison

CHAPTER 9

CONCLUSION AND FUTURE SCOPE

9.1. CONCLUSION AND FUTURE SCOPE

In conclusion, the process of performance comparison serves as a vital tool for evaluating and contrasting the effectiveness and efficiency of different systems, methods, or approaches based on various performance metrics. Through a systematic and structured evaluation process, organizations can gain valuable insights into the strengths and weaknesses of different solutions, enabling them to make informed decisions and optimize resources to achieve desired performance outcomes. By carefully selecting systems or methods to be evaluated, defining relevant performance metrics, and establishing an experimental setup that closely mimics real-world conditions, stakeholders can conduct comprehensive comparisons that yield actionable insights.

Throughout the performance comparison process, rigorous analysis and interpretation of data are essential to identify patterns, trends, and differences between the systems or methods being compared. Statistical analysis techniques may be employed to quantify the significance of observed differences and determine whether they are statistically meaningful. The findings of the comparison are then presented in a clear and understandable manner, typically through visualizations such as charts, graphs, or tables, accompanied by a detailed report summarizing the comparison methodology, results, and conclusions.

Looking ahead, there are several avenues for future research and exploration in the field of performance comparison. One potential area of focus is the development of more sophisticated performance metrics and evaluation methodologies that better capture the nuances and complexities

of modern systems and applications. Additionally, advancements in technology, such as the proliferation of cloud computing, edge computing, and artificial intelligence, present new opportunities and challenges for performance comparison. Future research may also explore the integration of performance comparison into the software development lifecycle, enabling continuous monitoring and optimization of system performance throughout the development process.

Furthermore, as organizations increasingly rely on data-driven decision-making, there is a growing need for tools and platforms that facilitate performance comparison and analysis. Future developments in this area may involve the creation of user-friendly interfaces, automated analysis workflows, and predictive modeling capabilities to support more efficient and effective performance comparison processes. Moreover, with the rise of interdisciplinary research and collaboration, there is potential for cross-pollination of ideas and methodologies from other fields, such as machine learning, optimization, and operations research, to further enhance the practice of performance comparison.

In conclusion, performance comparison plays a crucial role in evaluating and optimizing systems, methods, and approaches to achieve desired performance outcomes. By leveraging a systematic and structured approach to performance comparison, organizations can make informed decisions, drive continuous improvement, and ultimately, deliver better outcomes for stakeholders and end-users. As technology continues to evolve and new challenges emerge, the practice of performance comparison will remain essential for ensuring the effectiveness and efficiency of modern systems and applications.

APPENDIX – I

CODING

Program for Detection

File name as program.py

```
import cv2
import RPi.GPIO as GPIO
import time

# Set up GPIO mode and pins
GPIO.setmode(GPIO.BCM)
GPIO.setup(17, GPIO.OUT)
GPIO.setup(18, GPIO.OUT)
GPIO.setup(22, GPIO.OUT)
GPIO.setup(23, GPIO.OUT)
GPIO.setup(24, GPIO.OUT)
GPIO.setup(25, GPIO.OUT)

# Initialize PWM for motor speed control
pwm_motor1 = GPIO.PWM(24, 50) # ENA - Motor 1
speed control
pwm_motor2 = GPIO.PWM(25, 50) # ENB - Motor 2
speed control
pwm_motor1.start(0) # Start with 0% duty cycle
pwm_motor2.start(0)

# Define motor control functions
def stop():
    GPIO.output(17, GPIO.LOW)
    GPIO.output(18, GPIO.LOW)
    GPIO.output(22, GPIO.LOW)
    GPIO.output(23, GPIO.LOW)

def move_forward(speed=50):
    pwm_motor1.ChangeDutyCycle(min(speed, 50)) #
Limit speed to maximum of 50
    pwm_motor2.ChangeDutyCycle(min(speed, 50))
```

```

GPIO.output(17, GPIO.HIGH)
GPIO.output(18, GPIO.LOW)
GPIO.output(22, GPIO.HIGH)
GPIO.output(23, GPIO.LOW)

def turn_left(speed=50):
    pwm_motor1.ChangeDutyCycle(speed)
    pwm_motor2.ChangeDutyCycle(speed)
    GPIO.output(17, GPIO.LOW)
    GPIO.output(18, GPIO.HIGH)
    GPIO.output(22, GPIO.HIGH)
    GPIO.output(23, GPIO.LOW)

def turn_right(speed=50):
    pwm_motor1.ChangeDutyCycle(speed)
    pwm_motor2.ChangeDutyCycle(speed)
    GPIO.output(17, GPIO.HIGH)
    GPIO.output(18, GPIO.LOW)
    GPIO.output(22, GPIO.LOW)
    GPIO.output(23, GPIO.HIGH)

# Load pre-trained Haar Cascade classifier for pedestrian
detection
pedestrian_cascade =
cv2.CascadeClassifier(cv2.data.harcascades +
'haarcascade_frontalface_default.xml')

# Initialize video capture from the Pi camera (assuming
it's connected as '/dev/video0')
cap = cv2.VideoCapture(0)

# Define left and right lines for turning with increased
space
frame_width =
int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))
space_between_lines = 30 # Adjust as needed
line_left = frame_width // 3 - space_between_lines
line_right = frame_width * 2 // 3 + space_between_lines

```

```

# PID parameters
Kp = 0.1

# Proportional control gain

# Main loop

try:
    last_detection_position = None # Variable to store the
    last detected person's position

    while True:
        # Capture frame-by-frame
        ret, frame = cap.read()

        if not ret:
            break
        # Convert frame to grayscale for pedestrian detection
        gray = cv2.cvtColor(frame,
cv2.COLOR_BGR2GRAY)

        # Detect pedestrians in the frame
        pedestrians =
pedestrian_cascade.detectMultiScale(gray,
scaleFactor=1.1, minNeighbors=5, minSize=(30, 30))

        # Draw left and right lines
        cv2.line(frame, (line_left, 0), (line_left,
frame.shape[0]), (0, 255, 0), 2)
        cv2.line(frame, (line_right, 0), (line_right,
frame.shape[0]), (0, 255, 0), 2)

        person_detected = False

        # Process detected pedestrians
        for (x, y, w, h) in pedestrians:
            person_detected = True
            # Draw a rectangle around the detected face
            cv2.rectangle(frame, (x, y), (x+w, y+h), (255, 0,

```

0), 2)

```
# Calculate the center of the pedestrian bounding
box
pedestrian_center_x = x + w // 2

# Move robot based on pedestrian position

if line_left < pedestrian_center_x < line_right:
    # Move forward with dynamic speed adjustment
    based on distance
    speed = max(25, min(50, int(50 - (w /
frame_width) * 50)))
    # Adjust speed based on distance
    print("Move forward with speed:", speed)
    move_forward(speed)
    last_detection_position = 'forward'
else:
    # Adjust orientation using proportional control
    error = frame_width // 2 - pedestrian_center_x
    turn_rate = Kp * error
    if turn_rate > 0:
        print("Turn right")
        turn_right(int(turn_rate))
        last_detection_position = 'right'
    else:
        print("Turn left")
        turn_left(int(-turn_rate))
        last_detection_position = 'left'

# If no person detected
if not person_detected:
    print("No person detected")
    #stop()
    turn_left()
    time.sleep(0.3)

stop()
# Turn in the direction where the last person was
```

```

detected
    """if last_detection_position == 'forward':
        move_forward(25) # Move forward with
maximum speed
        elif last_detection_position == 'left':
            print("Turning left to find person")
            turn_left()
            time.sleep(0.5) # Adjust as needed
            move_forward(25)

# Move forward with maximum speed after turning left
        elif last_detection_position == 'right':
            print("Turning right to find person")
            turn_right()
            time.sleep(0.5) # Adjust as needed
            move_forward(25) # Move forward with
maximum speed after turning right
        """

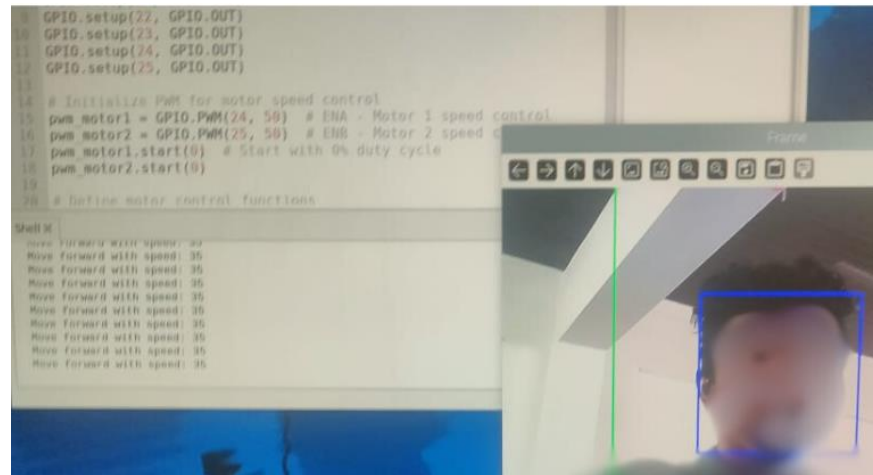
        # Display the resulting frame
        cv2.imshow('Frame', frame)

        # Break the loop if 'q' is pressed
        if cv2.waitKey(1) & 0xFF == ord('q'):
            break
finally:
    # Release the capture
    cap.release()
    cv2.destroyAllWindows()
    # Stop motors and cleanup GPIO pins
    stop()
    pwm_motor1.stop()
    GPIO.cleanup()

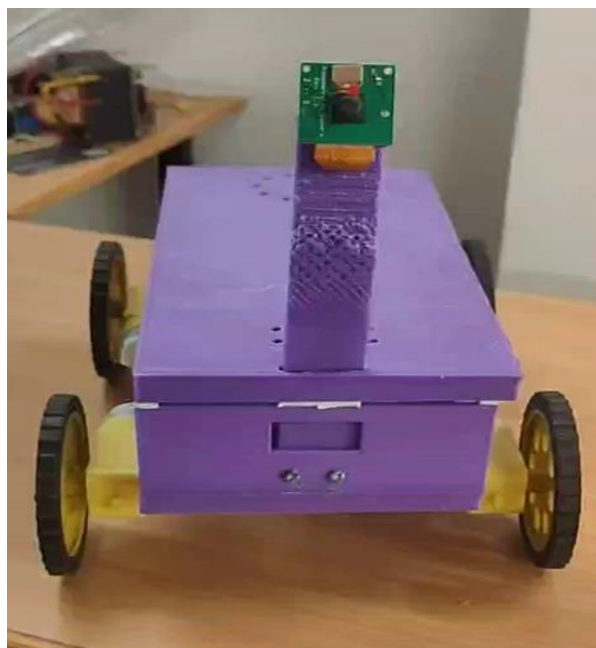
```


APPENDIX – II

OUTPUT:



OUTPUT 1: In Thonny of Raspberry Pi



OUTPUT2 : Product

APPENDIX – III

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